Faculty of Engineering Computer Engineering Department
Artificial Intelligence
ENCS3340
Project#1

Optimization Strategies for Local Package Delivery Operations

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Section : 3

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Abstract

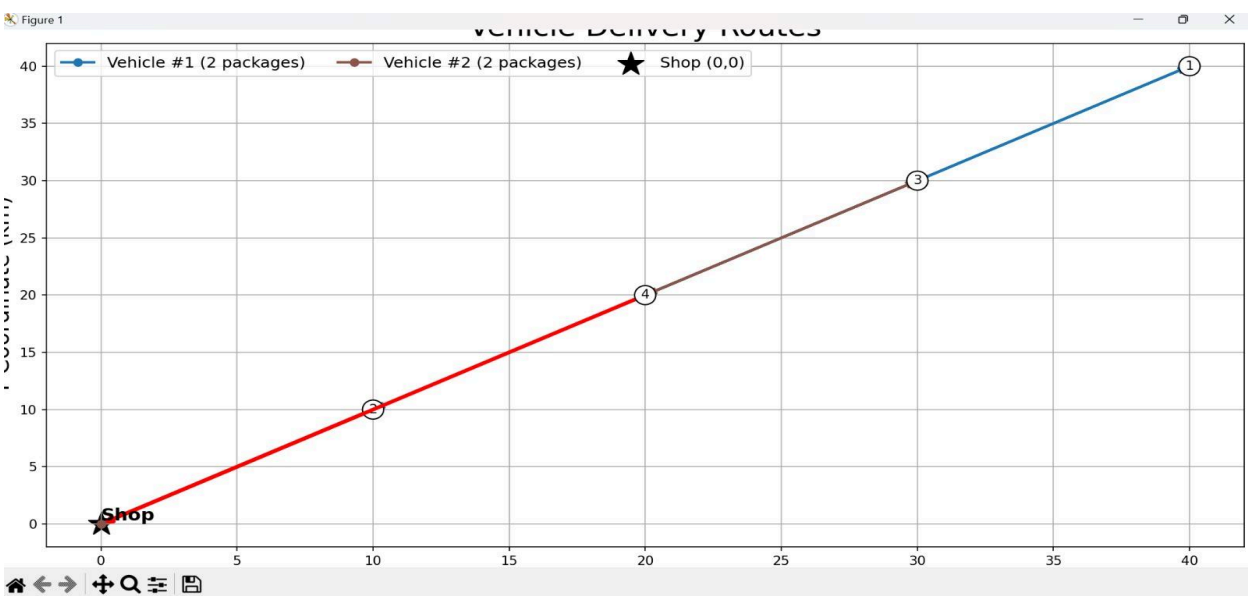
Our project suggests an optimization model for local package delivery operations by metaheuristic algorithms. The primary objective is to route the packages into vehicles such that the total distance traveled is minimized, with no vehicle at or above capacity. Packages are described by destination coordinates, weight, and priority. Two optimization approaches, Simulated Annealing (SA) and Genetic Algorithm (GA), were used. The program receives user input on package information and vehicle capacities and plots the routes of delivery in 2D. Results demonstrate that Simulated Annealing suffers from cost minimization and fast convergence, while the Genetic Algorithm requires careful setting to avoid the local optima. Both schemes were capable of generating feasible capacity-compliant assignments. This paper demonstrates the capability of search-based optimization techniques as applied to practical logistics problems.

Test Cases:

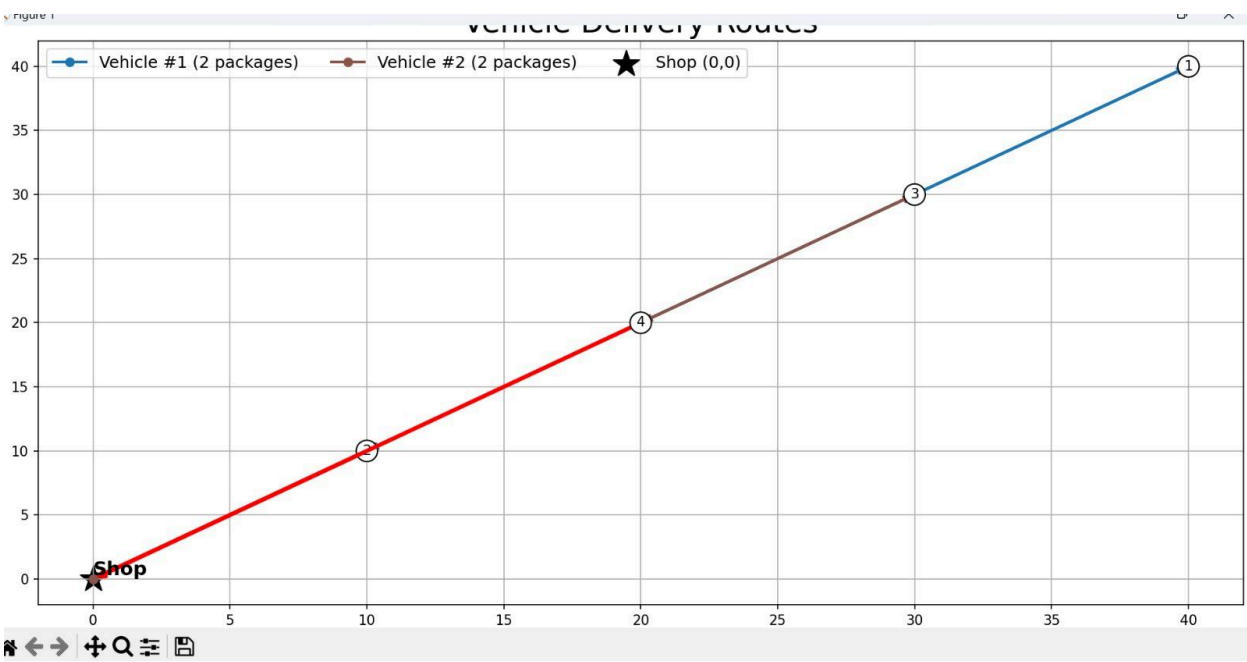
1.Simple Test Case

```
===== Vehicle and Package Setup =====
Enter the number of available vehicles: 2
Enter capacity of vehicle #1 (kg): 100
Enter capacity of vehicle #2 (kg): 100

Enter the number of packages: 4

Package #1:
  - X coordinate: 10
  - Y coordinate: 10
  - Weight (kg): 30
  - Priority (1=highest): 2

Package #2:
  - X coordinate: 20
  - Y coordinate: 20
  - Weight (kg): 40
  - Priority (1=highest): 2

Package #3:
  - X coordinate: 30
  - Y coordinate: 30
  - Weight (kg): 50
  - Priority (1=highest): 2

Package #4:
  - X coordinate: 40
  - Y coordinate: 40
  - Weight (kg): 60
  - Priority (1=highest): 2
```

```
========= Algorithm Selection =========
Choose an optimization algorithm:
1. Simulated Annealing (SA)
2. Genetic Algorithm (GA)
Enter your choice (1 or 2): 1

Starting Simulated Annealing...
Initial solution cost: 197.99 (0 unassigned)

=== Simulated Annealing Results ===
Final cost: 197.99

=== Assignment Summary ===
Total distance: 197.99 km
Unassigned packages: 0

--- Vehicle Details ---

Vehicle #1 (Capacity: 90.00/100.0 kg):
  - Package(dest=[40.0,40.0], weight=60.0, priority=2)
  - Package(dest=[10.0,10.0], weight=30.0, priority=2)

Vehicle #2 (Capacity: 90.00/100.0 kg):
  - Package(dest=[30.0,30.0], weight=50.0, priority=2)
  - Package(dest=[20.0,20.0], weight=40.0, priority=2)
```



```
========= Algorithm Selection =========
Choose an optimization algorithm:
1. Simulated Annealing (SA)
2. Genetic Algorithm (GA)
Enter your choice (1 or 2): 2

Starting Genetic Algorithm...
Generation 0: Best fitness = 197.99 (0 unassigned)
Generation 50: Best fitness = 197.99 (0 unassigned)
Generation 100: Best fitness = 197.99 (0 unassigned)
Generation 150: Best fitness = 197.99 (0 unassigned)
Generation 200: Best fitness = 197.99 (0 unassigned)
Generation 250: Best fitness = 197.99 (0 unassigned)
Generation 300: Best fitness = 197.99 (0 unassigned)
Generation 350: Best fitness = 197.99 (0 unassigned)
Generation 400: Best fitness = 197.99 (0 unassigned)
Generation 450: Best fitness = 197.99 (0 unassigned)

=== Genetic Algorithm Results ===
Final fitness: 197.99

=== Assignment Summary ===
Total distance: 197.99 km
Unassigned packages: 0

--- Vehicle Details ---

Vehicle #1 (Capacity: 90.00/100.0 kg):
  - Package(dest=[40.0,40.0], weight=60.0, priority=2)
  - Package(dest=[10.0,10.0], weight=30.0, priority=2)

Vehicle #2 (Capacity: 90.00/100.0 kg):
  - Package(dest=[30.0,30.0], weight=50.0, priority=2)
  - Package(dest=[20.0,20.0], weight=40.0, priority=2)
```



Results:

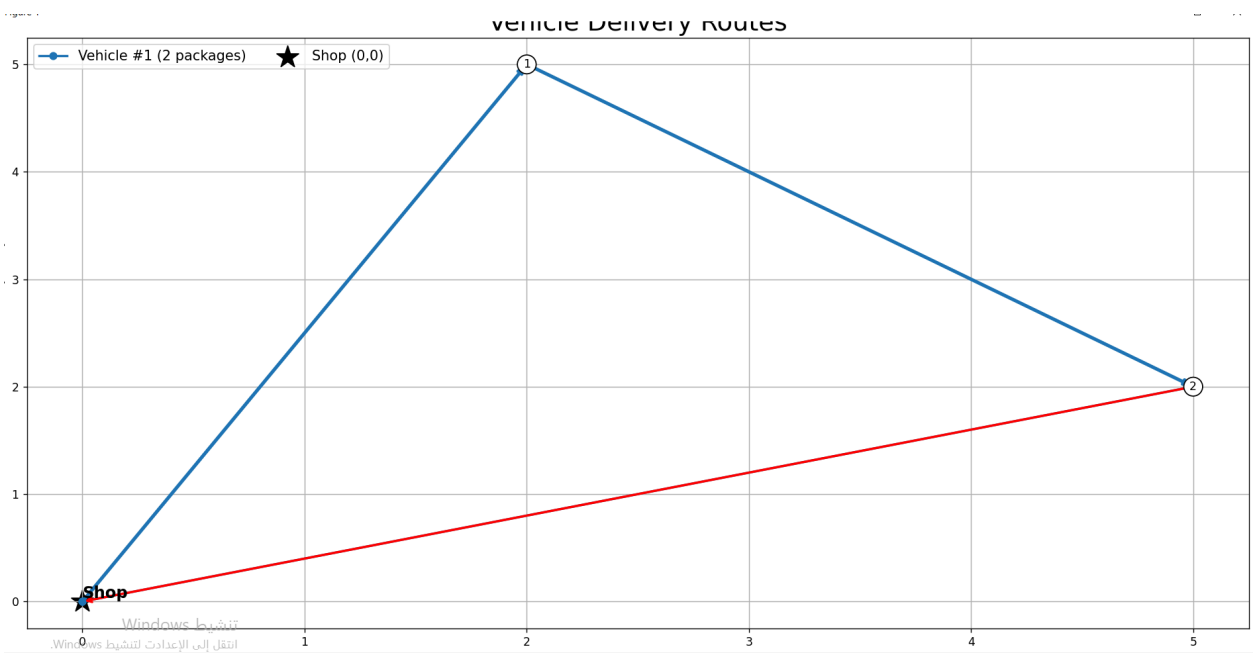
1. All packages assigned successfully to vehicles (0 unassigned).
2. Each vehicle used 90/100 kg capacity – efficient usage.
3. Total distance traveled: 197.99 km (same for both GA and SA).

Both the Genetic Algorithm (GA) and Simulated Annealing (SA) were successfully executed and produced identical results: all packages were assigned without exceeding vehicle capacities, and the total delivery distance was 197.99 km. Since all packages had the same priority level, the priority factor had no influence on the outcome. From a technical perspective, GA is generally more suitable for complex, multi-constraint problems, offering broader exploration, while SA tends to be faster but may settle on suboptimal solutions.

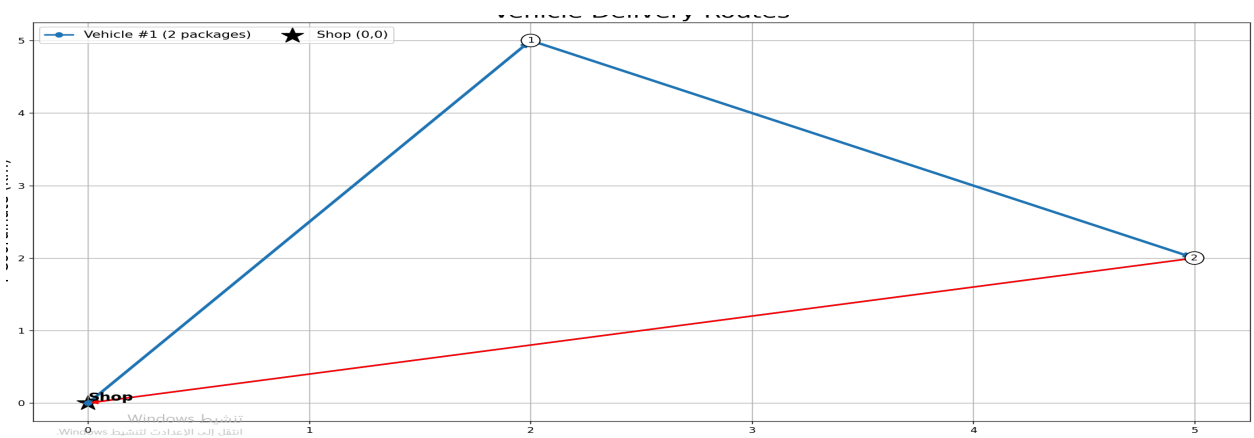
2.

```
===== Vehicle and Package Setup =====
Enter the number of available vehicles: 1
Enter capacity of vehicle #1 (kg): 100

Enter the number of packages: 3

Package #1:
  - X coordinate: 2
  - Y coordinate: 5
  - Weight (kg): 80
  - Priority (1=highest): 1

Package #2:
  - X coordinate: 5
  - Y coordinate: 7
  - Weight (kg): 30
  - Priority (1=highest): 3

Package #3:
  - X coordinate: 5
  - Y coordinate: 2
  - Weight (kg): 10
  - Priority (1=highest): 2
```

```
========== Algorithm Selection ==========
Choose an optimization algorithm:
1. Simulated Annealing (SA)
2. Genetic Algorithm (GA)
Enter your choice (1 or 2): 1

Starting Simulated Annealing...
Initial solution cost: 1015.01 (1 unassigned)

=== Simulated Annealing Results ===
Final cost: 1015.01

=== Assignment Summary ===
Total distance: 15.01 km
Unassigned packages: 1

--- Vehicle Details ---

Vehicle #1 (Capacity: 90.00/100.0 kg):
  - Package(dest=[2.0,5.0], weight=80.0, priority=1)
  - Package(dest=[5.0,2.0], weight=10.0, priority=2)

--- Unassigned Packages ---
  - Package(dest=[5.0,7.0], weight=30.0, priority=3)
```



```
========= Algorithm Selection =========
Choose an optimization algorithm:
1. Simulated Annealing (SA)
2. Genetic Algorithm (GA)
Enter your choice (1 or 2): 2

Starting Genetic Algorithm...
Generation 0: Best fitness = 1015.01 (1 unassigned)
Generation 50: Best fitness = 1015.01 (1 unassigned)
Generation 100: Best fitness = 1015.01 (1 unassigned)
Generation 150: Best fitness = 1015.01 (1 unassigned)
Generation 200: Best fitness = 1015.01 (1 unassigned)
Generation 250: Best fitness = 1015.01 (1 unassigned)
Generation 300: Best fitness = 1015.01 (1 unassigned)
Generation 350: Best fitness = 1015.01 (1 unassigned)
Generation 400: Best fitness = 1015.01 (1 unassigned)
Generation 450: Best fitness = 1015.01 (1 unassigned)

=== Genetic Algorithm Results ===
Final fitness: 1015.01

=== Assignment Summary ===
Total distance: 15.01 km
Unassigned packages: 1

--- Vehicle Details ---

Vehicle #1 (Capacity: 90.00/100.0 kg):
  - Package(dest=[2.0,5.0], weight=80.0, priority=1)
  - Package(dest=[5.0,2.0], weight=10.0, priority=2)

--- Unassigned Packages ---
  - Package(dest=[5.0,7.0], weight=30.0, priority=3)
```



Note the red line represent the return route to the shop

Results:

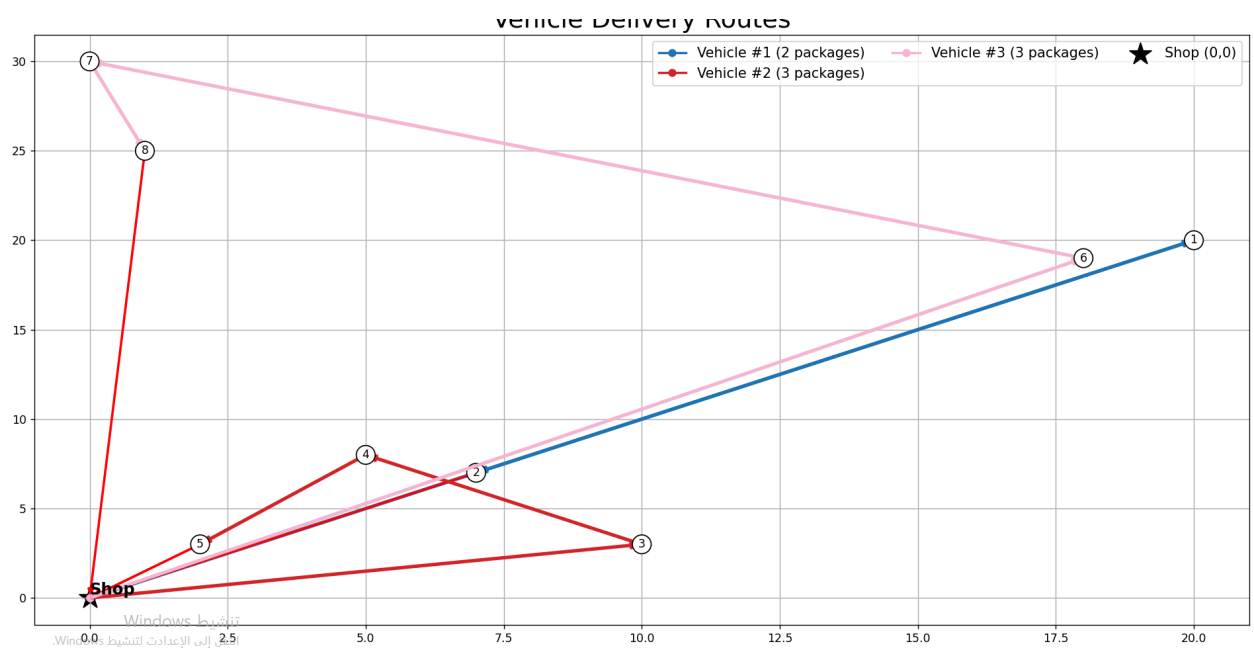
Total distance traveled: 15.01 km (same for both GA and SA).
Both algorithms performed similarly, with Genetic Algorithm and Simulated Annealing both leading to the same final solution. However, both left one package unassigned, primarily due to the weight capacity constraints. Neither algorithm demonstrated significant improvements in cost, suggesting that further parameter tuning, better solution diversity, or additional vehicles might be needed to achieve a more optimal result.

3.

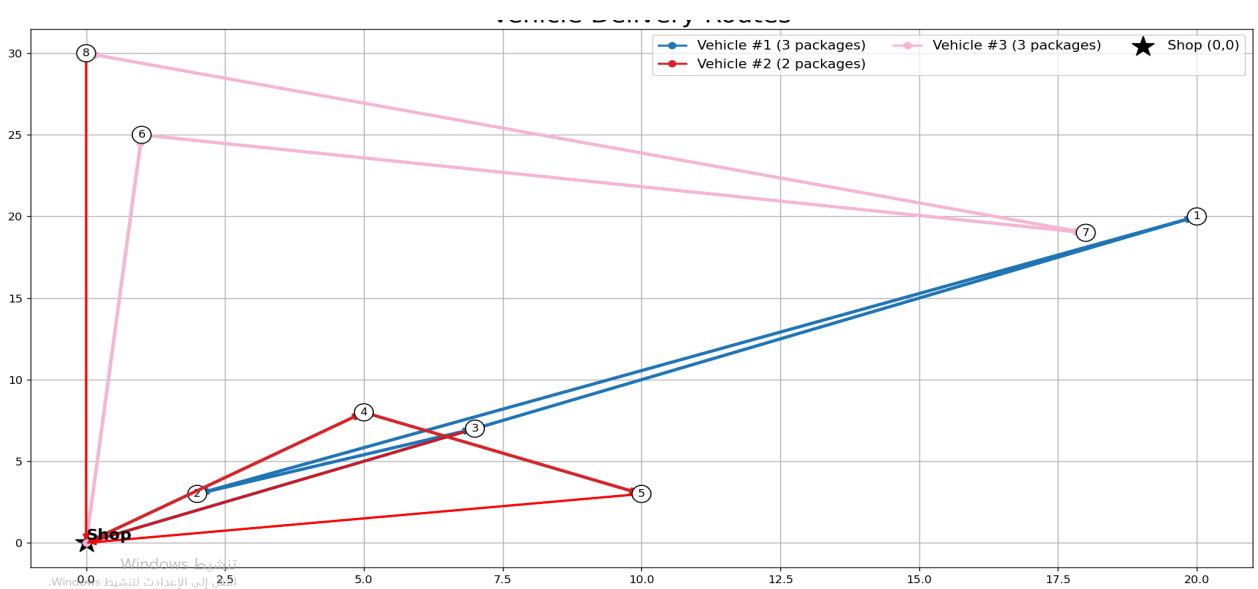
```
===== Vehicle and Package Setup =====
Enter the number of available vehicles: 3
Enter capacity of vehicle #1 (kg): 20
Enter capacity of vehicle #2 (kg): 20
Enter capacity of vehicle #3 (kg): 20

Enter the number of packages: 8

Package #1:
  - X coordinate: 2
  - Y coordinate: 3
  - Weight (kg): 5
  - Priority (1=highest): 2

Package #2:
  - X coordinate: 5
  - Y coordinate: 8
  - Weight (kg): 7
  - Priority (1=highest): 1

Package #3:
  - X coordinate: 10
  - Y coordinate: 3
  - Weight (kg): 8
  - Priority (1=highest): 2

Package #4:
  - X coordinate: 7
  - Y coordinate: 7
  - Weight (kg): 6
  - Priority (1=highest): 3

Package #5:
  - X coordinate: 20
  - Y coordinate: 20
  - Weight (kg): 9
  - Priority (1=highest): 1

Package #6:
```

```
Package #6:
  - X coordinate: 18
  - Y coordinate: 19
  - Weight (kg): 10
  - Priority (1=highest): 2

Package #7:
  - X coordinate: 0
  - Y coordinate: 30
  - Weight (kg): 6
  - Priority (1=highest): 3

Package #8:
  - X coordinate: 1
  - Y coordinate: 25
  - Weight (kg): 4
  - Priority (1=highest): 1

========= Algorithm Selection =========
Choose an optimization algorithm:
1. Simulated Annealing (SA)
2. Genetic Algorithm (GA)
Enter your choice (1 or 2): 1
```

```
Starting Simulated Annealing...
Initial solution cost: 190.43 (0 unassigned)
Iter 0: New best cost: 177.77 (0 unassigned)
Iter 0: New best cost: 161.01 (0 unassigned)
Iter 0: New best cost: 160.98 (0 unassigned)
Iter 0: New best cost: 160.90 (0 unassigned)

=== Simulated Annealing Results ===
Final cost: 160.90

=== Assignment Summary ===
Total distance: 160.90 km
Unassigned packages: 0

--- Vehicle Details ---

Vehicle #1 (Capacity: 15.00/20.0 kg):
  - Package(dest=[20.0,20.0], weight=9.0, priority=1)
  - Package(dest=[7.0,7.0], weight=6.0, priority=3)

Vehicle #2 (Capacity: 20.00/20.0 kg):
  - Package(dest=[10.0,3.0], weight=8.0, priority=2)
  - Package(dest=[5.0,8.0], weight=7.0, priority=1)
  - Package(dest=[2.0,3.0], weight=5.0, priority=2)

Vehicle #3 (Capacity: 20.00/20.0 kg):
  - Package(dest=[18.0,19.0], weight=10.0, priority=2)
  - Package(dest=[0.0,30.0], weight=6.0, priority=3)
  - Package(dest=[1.0,25.0], weight=4.0, priority=1)
```



```
========== Algorithm Selection ==========
Choose an optimization algorithm:
1. Simulated Annealing (SA)
2. Genetic Algorithm (GA)
Enter your choice (1 or 2): 2

Starting Genetic Algorithm...
Generation 0: Best fitness = 190.43 (0 unassigned)
Generation 50: Best fitness = 190.43 (0 unassigned)
Generation 100: Best fitness = 190.43 (0 unassigned)
Generation 150: Best fitness = 190.43 (0 unassigned)
Generation 200: Best fitness = 190.43 (0 unassigned)
Generation 250: Best fitness = 190.43 (0 unassigned)
Generation 300: Best fitness = 190.43 (0 unassigned)
Generation 350: Best fitness = 190.43 (0 unassigned)
Generation 400: Best fitness = 190.43 (0 unassigned)
Generation 450: Best fitness = 190.43 (0 unassigned)

=== Genetic Algorithm Results ===
Final fitness: 190.43

=== Assignment Summary ===
Total distance: 190.43 km
Unassigned packages: 0

--- Vehicle Details ---

Vehicle #1 (Capacity: 20.00/20.0 kg):
  - Package(dest=[20.0,20.0], weight=9.0, priority=1)
  - Package(dest=[2.0,3.0], weight=5.0, priority=2)
  - Package(dest=[7.0,7.0], weight=6.0, priority=3)

Vehicle #2 (Capacity: 15.00/20.0 kg):
  - Package(dest=[5.0,8.0], weight=7.0, priority=1)
  - Package(dest=[10.0,3.0], weight=8.0, priority=2)

Vehicle #3 (Capacity: 20.00/20.0 kg):
  - Package(dest=[1.0,25.0], weight=4.0, priority=1)
  - Package(dest=[18.0,19.0], weight=10.0, priority=2)
  - Package(dest=[0.0,30.0], weight=6.0, priority=3)
```



Note:the red line represent the return route to the shop

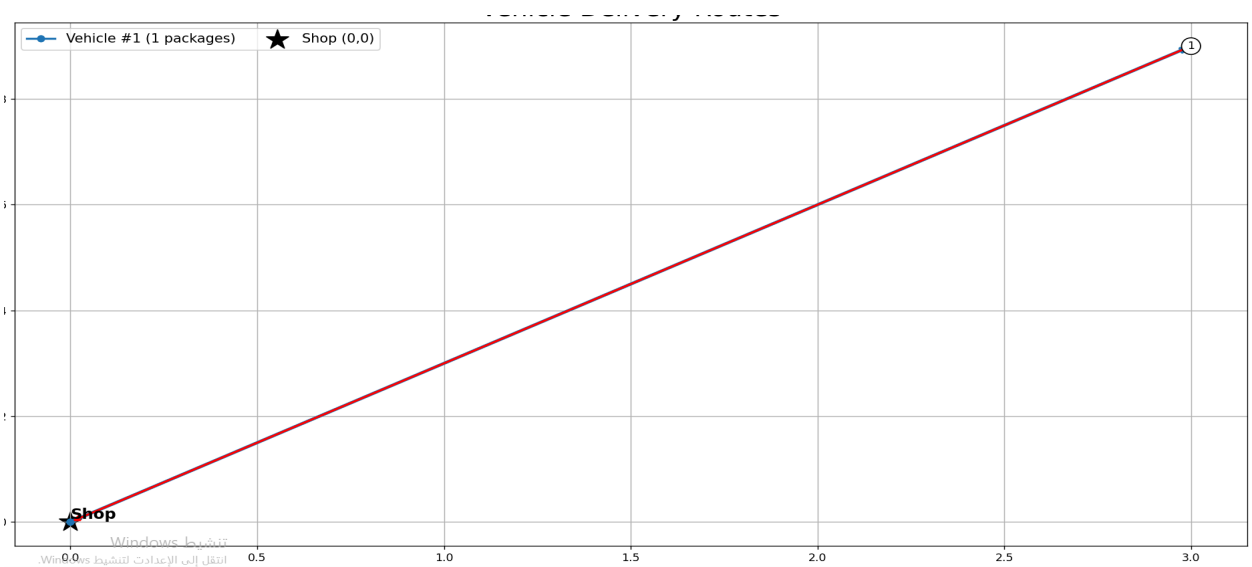
Results:

Simulated Annealing (SA) obtained a better solution at a cost of 160.90 km towards the later end, while Genetic Algorithm (GA) was at a cost of 190.43 km.Both algorithms managed to assign all packages, and there were 0 packages not assigned.Simulated Annealing reduced its cost from generation to generation, which indicates improving search in the solution space. Genetic Algorithm did not improve after the first generation, which indicates stagnation.In general, Simulated Annealing performed better than Genetic Algorithm both in solution quality (lower total distance) and convergence speed.

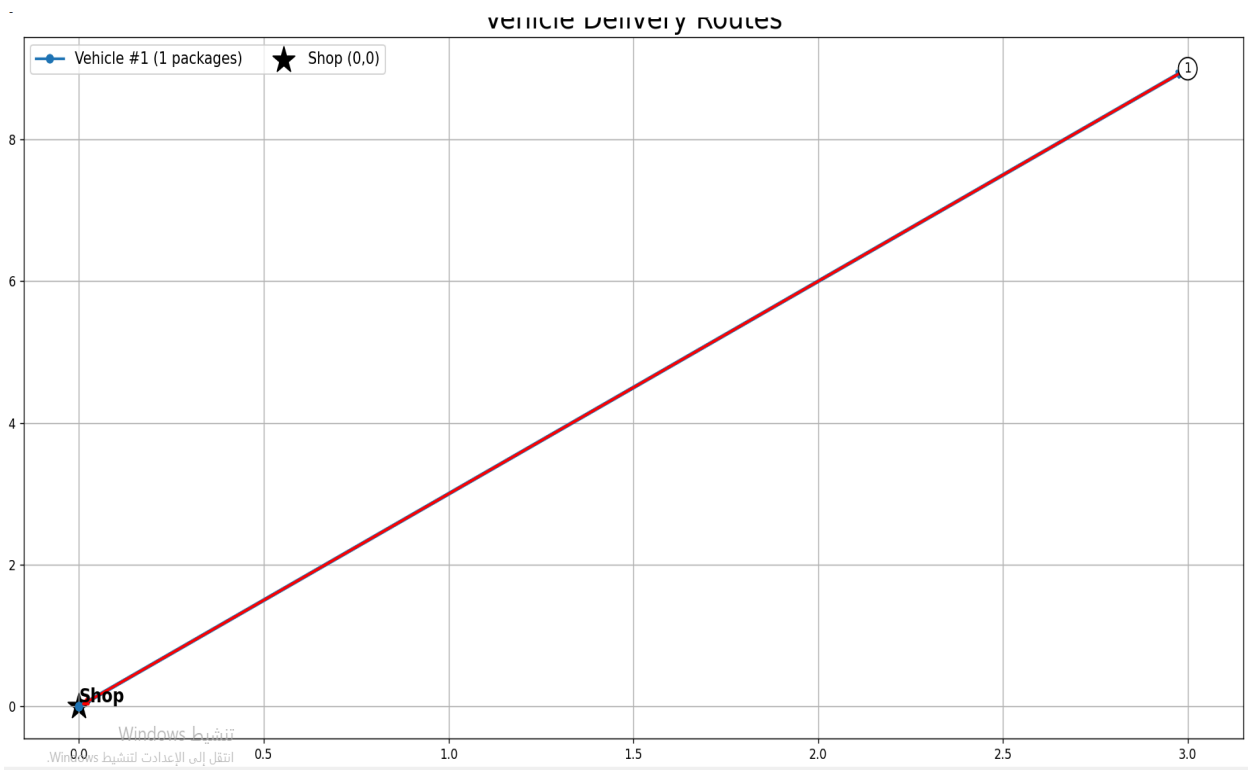
4.

```
===== Vehicle and Package Setup =====
Enter the number of available vehicles: 1
Enter capacity of vehicle #1 (kg): 100

Enter the number of packages: 2

Package #1:
  - X coordinate: 4
  - Y coordinate: 5
  - Weight (kg): 90
  - Priority (1=highest): 4

Package #2:
  - X coordinate: 3
  - Y coordinate: 9
  - Weight (kg): 20
  - Priority (1=highest): 1

========= Algorithm Selection =========
Choose an optimization algorithm:
1. Simulated Annealing (SA)
2. Genetic Algorithm (GA)
Enter your choice (1 or 2): 1
```

```
========= Algorithm Selection =========
Choose an optimization algorithm:
1. Simulated Annealing (SA)
2. Genetic Algorithm (GA)
Enter your choice (1 or 2): 1

Starting Simulated Annealing...
Initial solution cost: 1018.97 (1 unassigned)

=== Simulated Annealing Results ===
Final cost: 1018.97

=== Assignment Summary ===
Total distance: 18.97 km
Unassigned packages: 1

--- Vehicle Details ---

Vehicle #1 (Capacity: 20.00/100.0 kg):
  - Package(dest=[3.0,9.0], weight=20.0, priority=1)

--- Unassigned Packages ---
  - Package(dest=[4.0,5.0], weight=90.0, priority=4)
```



```
========= Algorithm Selection =========
Choose an optimization algorithm:
1. Simulated Annealing (SA)
2. Genetic Algorithm (GA)
Enter your choice (1 or 2): 2

Starting Genetic Algorithm...
Generation 0: Best fitness = 1018.97 (1 unassigned)
Generation 50: Best fitness = 1018.97 (1 unassigned)
Generation 100: Best fitness = 1018.97 (1 unassigned)
Generation 150: Best fitness = 1018.97 (1 unassigned)
Generation 200: Best fitness = 1018.97 (1 unassigned)
Generation 250: Best fitness = 1018.97 (1 unassigned)
Generation 300: Best fitness = 1018.97 (1 unassigned)
Generation 350: Best fitness = 1018.97 (1 unassigned)
Generation 400: Best fitness = 1018.97 (1 unassigned)
Generation 450: Best fitness = 1018.97 (1 unassigned)

=== Genetic Algorithm Results ===
Final fitness: 1018.97

=== Assignment Summary ===
Total distance: 18.97 km
Unassigned packages: 1

--- Vehicle Details ---

Vehicle #1 (Capacity: 20.00/100.0 kg):
  - Package(dest=[3.0,9.0], weight=20.0, priority=1)

--- Unassigned Packages ---
  - Package(dest=[4.0,5.0], weight=90.0, priority=4)
PS C:\Users\shahed\Desktop\lec1>
```



Results:

In both instances, the Genetic Algorithm (GA) and Simulated Annealing (SA) both yielded the same outcome: one of the two packages (90 kg, priority 4) was left out, and the lighter but more urgent package (20 kg, priority 1) was delivered. This outcome verifies that the two algorithms both respected the vehicle capacity constraint (100 kg) and both respected package prioritization by urgency, with urgent deliveries being prioritized under low capacity. In the performance, SA worked better by converging to the optimal solution quickly, while GA was not able to produce a better configuration even after 450 generations. It means that for small-sized instances of the problem, SA can work more effectively and efficiently than GA, particularly if high-priority constraints are to be satisfied under constrained capacity.

Conclusion

In general, the project successfully showed how Simulated Annealing (SA) and Genetic Algorithm (GA) can be applied to optimize local package delivery operations under real-world constraints such as vehicle capacity, package weight, and priority delivery. The system dynamically assigned packages to vehicles and generated efficient delivery routes that minimize overall travel distance while honoring priority deliveries. Using a set of test cases, it was observed that SA tends to perform better than GA for smaller problem instances in terms of time and solution quality, especially where handling priorities is crucial. The project illustrates how optimization of algorithms can significantly enhance logistical efficiency and provides a good foundation for scaling up to more intricate, real-world delivery networks.